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Acomys, the closest relatives to gerbils, do express Pdx-1 protein and have similar islet morphology to gerbils

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Keywords: *Acomys*, Pdx-1, transcription factors, MafA, MafB, Nkx6.1, Nkx2.2, glucagon, insulin, somatostatin, gerbil, spiny mouse

Abbreviations: SS, somatostatin; PP, pancreatic polypeptide; CART, cocaine amphetamine regulated transcript; CCK, cholecystokinin; IAPP, islet amyloid polypeptide; PYY, peptide YY; TRH, thyrotropin releasing hormone; MafA, v-Maf musculoaponeurotic fibrosarcoma oncogene homolog A; MafB, v-Maf musculoaponeurotic fibrosarcoma oncogene homolog B; Pdx-1, pancreatic duodenal homeobox-1; Nkx2.2, NK-related homeobox 2.2; Nkx6.1, NK-related homeobox 6.1; Isl-1, LIM domain homeobox gene islet 1; TSA, tyramide amplification system; PCR, polymerase chain reaction; RACE, rapid amplification of cDNA ends; HE, high energy; IpF-1, insulin promoter factor-1

Acomys, also called spiny mice, were once used as a diabetes model. We have recently demonstrated that the closest relatives to the *Acomys*, members of the family Gerbillinae, lack the transcription factor Pdx-1. Therefore, we sought to determine if members of this family also lack Pdx-1, and describe the pancreatic morphology in three different species of *Acomys*: *Acomys cahirinus* (Egyptian spiny mouse), *Acomys cilicicus* (Asia Minor spiny mouse) and *Acomys dimidiatus* (eastern spiny mouse). We successfully cloned the *Acomys* Pdx-1 gene and we demonstrate by immunocytochemistry that the Pdx-1 protein is expressed in the pancreatic insulin immunoreactive cells and in a subset of the somatostatin cells. The basic islet structure is very similar to other rodents—with the insulin cells in the center, and glucagon, somatostatin, PP and occasional PYY cells in the periphery. No ghrelin or CART cells were identified. Nkx6.1 was localized specifically to the insulin immunoreactive cells, while Nkx2.2 was found in all endocrine cells except the somatostatin immunoreactive cells. Both MafA and MafB were expressed in the islets; MafA being specific for the insulin cells, while MafB was primarily in the glucagon cells but also found in some insulin cells. Isl-1 was localized in all endocrine cell types. In conclusion, the closest relatives to the gerbils express a Pdx-1 protein that is 90% similar to other rodents but also has a unique 3 amino acid insert compared to other species. During the evolution of the spiny mice and the gerbils, it appears that the Pdx-1 gene was lost.

Introduction

The pancreatic islets of Langerhans consist of four other major cell types: the insulin producing β -cells, the α -cells which produce glucagon, δ -cells producing somatostatin (SS) and PP-cells producing pancreatic polypeptide (PP). In addition, in some species, other peptide producing cells can be found in the islets of Langerhans. These include cocaine amphetamine regulated transcript (CART), cholecystokinin (CCK), ghrelin (ϵ -cells), islet amyloid polypeptide (IAPP), peptide YY (PYY) and thyrotropin releasing hormone (TRH).¹⁻⁷

Recent discoveries have shown that numerous transcription factors operate in the pancreas, both during development and in the later function and maintenance of the different cell types in the adult pancreas.^{8,9} One of the most important

is pancreatic duodenal homeobox-1 (Pdx-1) which plays a vital role as demonstrated by a complete lack of the pancreas organ in its absence,^{10,11} whereas deletion of Pdx-1 expression in adult mice leads to diabetes.¹² We reported that the Pdx-1 gene cannot be detected in most of species belonging to the subfamily Gerbillinae.¹³ This fact has lead us to further study the evolution of the Pdx-1 gene. All so far examined members of the family Muridae do express Pdx-1. One subfamily of Muridae, Leimacomyinae, is thought to be extinct, therefore it has never been examined. We have recently reported that one member of the subfamily Otomyinae, *Otomys sloggetti robertsi* (African Ice Rat), expressed the Pdx-1 protein in the pancreatic β -cells.¹⁴ Members of the subfamily Deomyinae, the closest relatives to the Gerbillinae,¹⁵ have never been examined for the expression of Pdx1.

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Table 1. Percentage similarity among different species

	Nucleotide	Protein
<i>Acomys cahirinus</i>	100%	100%
<i>Rattus norvegicus</i>	90.0%	93.4%
<i>Mus musculus</i>	89.5%	91.6%
<i>Homo sapiens</i>	86.4%	88.5%

The genus *Acomys* belongs to the subfamily Deomyinae, family Muridae. Among the 19 species in this genus, we studied *Acomys cahirinus* (Egyptian spiny mouse), *Acomys cilicicus* (Asia Minor spiny mouse) and *Acomys dimidiatus* (Eastern spiny mouse). As their common name suggests, the *Acomys* group was traditionally classified within the subfamily Murinae (old world mice and rats).^{16,17} *Acomys* species live in the arid areas of eastern Mediterranean countries and in North Africa. The animals are nocturnal and have a light brown fur with bristles on their backs—hence “spiny” mice.¹⁸ Coincidence was part of the reason that *Acomys* species (spiny mice) were an animal model used in

the study of diabetes (diabetes + obesity). Overweight, glucosuria and ketosis were discovered when animals were kept as pets feeding on bird food rich in pumpkin, sesame and sunflower seeds.^{19,20} Later studies of the animals established that spiny mice demonstrate low insulin secretion capacity and a decreased response to glucose even with islet hypertrophy and hyperplasia.^{21–25}

As some *Acomys* species have a tendency to become diabetic on a laboratory diet similar to their closest relatives, the gerbils, we hypothesized that they might also lack the expression of Pdx-1. In addition, we have investigated the morphology and expression of the major hormones and transcription factors.

Results

Pdx-1 is expressed at the gene and protein level in *Acomys*. *Acomys* are the closest relatives to the Gerbillinae, that were reported to lack the Pdx-1.¹³ Additionally, *Acomys* have been reported to have defective insulin secretion and to easily develop diabetes on laboratory diets, thus we hypothesized that perhaps, like the gerbils,

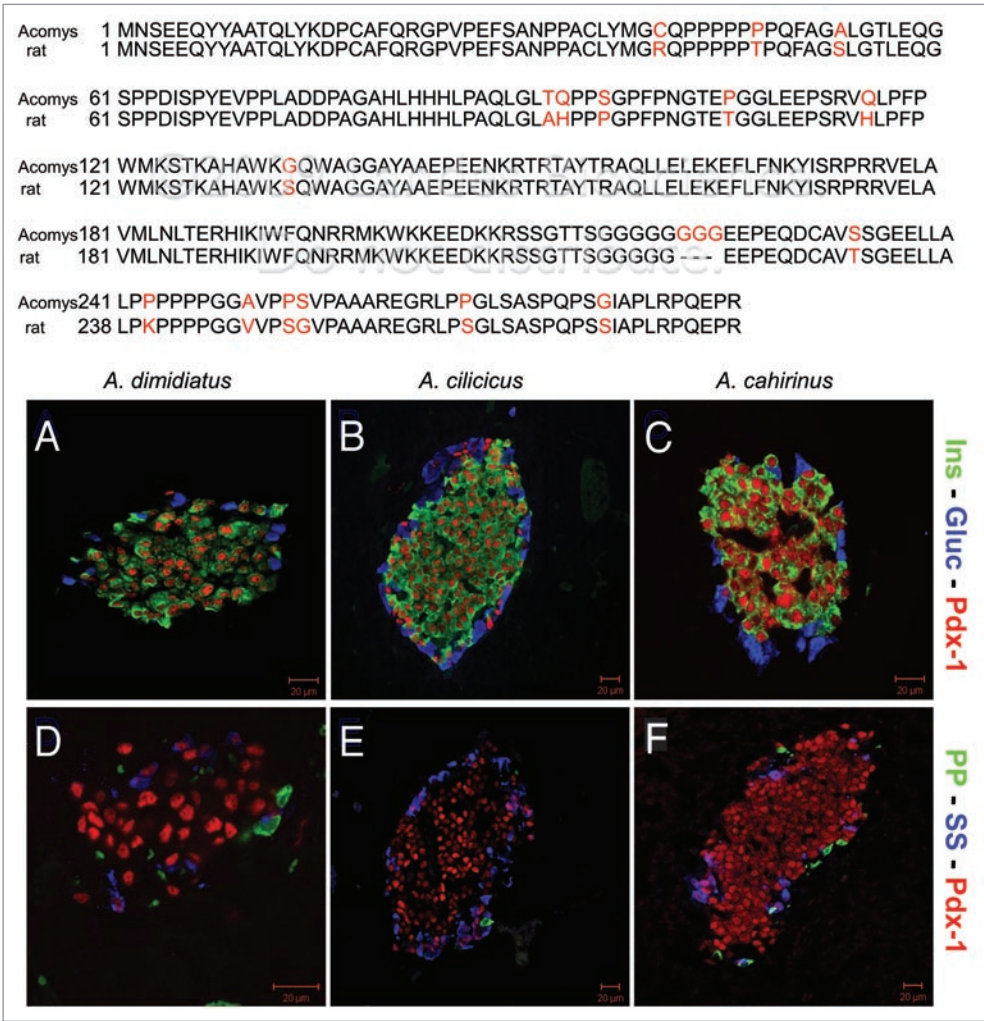


Figure 1. The amino acid alignment of the rat and *Acomys cahirinus* Pdx-1 protein. Amino acid changes are highlighted in red. Immunoreactivity for Pdx-1 in three different species of *Acomys*. (A–C) Immunoreactivity for insulin (green), glucagon (blue) and Pdx-1 (red) in three species of *Acomys*. (D–F) Immunoreactivity for pancreatic polypeptide (green), somatostatin (blue) and Pdx-1 (red) in three species of *Acomys*. Scale bar = 20 μm.

this subfamily might also lack Pdx-1. We examined both by PCR and immunostaining for the presence of Pdx-1 in three species of *Acomys*. Using our conserved Pdx-1 primers¹³ and other primers, we successfully cloned the full length coding sequence of the gene. The alignment of the nucleotide sequence demonstrates a 90% similarity to rats, 89.5% to mice, 89% to hamster, 86.4% to humans and 60.3% to *Xenopus* (Table 1). Interestingly, there are 16 different amino acids plus three additional glycines extending the glycine rich stretch to 8 amino acids (Fig. 1). Five of the 16 amino acids are surprisingly changed to proline. The homeodomain from amino acids 147 to 205 is 100% conserved as would be expected. Pdx-1 stained the nuclei of the insulin immunoreactive cells and also a sub-population of the somatostatin immunoreactive cells in all three *Acomys* species, whereas neither of the glucagon nor PP-cells stained for Pdx-1 (Fig. 1). Thus, the most closely related animals to gerbils do express the Pdx-1 gene and protein.

Acomys show a normal hormonal localization in islets. For the first time, we observed that spiny mice have islets varying in size. Few very large islets were often found with mostly smaller islets distributed in the exocrine tissue. Also small groups or single endocrine cells were found scattered in the exocrine pancreas. The distribution of five different hormones—insulin, glucagon, somatostatin, pancreatic polypeptide and ghrelin—was examined in the pancreas of the three *Acomys* species. In all three species, the insulin producing β -cells represented the majority of the endocrine cells and these were placed in the center of the islets (Fig. 2A–F). Mainly surrounding the β -cells were the α -cells producing glucagon (Fig. 2A–F). These were found in a significantly lower number than the β -cells, and in some islets they were as few as either SS-cells or PP-cells, which were found scattered in the periphery of the islets (Fig. 2A–F). Peptide YY (PYY) is a hormone found to be expressed in some rodent pancreata.^{24,25} We had previously demonstrated that both PYY and CART were localized in the islets of the African Ice Rat.¹⁴ We observed PYY in all species and found it to be expressed in a few cells within some islets (data not shown). Ghrelin and CART expressing cells were not detected in any of the species (data not shown).

Nkx6.1, Nkx2.2, MafA, MafB and Isl-1 are all expressed in Acomys islets. Nkx6.1 was found to be expressed exclusively in the insulin immunoreactive cells, while Nkx2.2 was expressed at variable levels in all the hormone producing cells except for the somatostatin immunoreactive cells (Fig. 3). MafA was expressed solely in the insulin immunoreactive cells, however some insulin cells showed little or no MafA immunoreactivity (Fig. 4). MafB was strongly immunoreactive in most of the glucagon immunoreactive cells and interestingly, MafB was expressed at different levels in some of the insulin immunoreactive cells. Neither PP nor somatostatin cells were found to express MafB. Isl-1/2 immunoreactivity was found in all the endocrine cells except for a few insulin cells (data not shown).

Discussion

The homeodomain transcription factor Pdx-1 is essential for pancreas development, insulin production and glucose homeostasis.²⁶ Mutations in Pdx-1(Ipf-1) are associated both with

Table 2. Antisera/Antibodies used in this study.

Antigen	Species	Source	Dilution
Glucagon	Guinea Pig	Linco	1:500
Glucagon	Mouse	Novo Nordisk	1:50
Glucagon	Rabbit	DAKO	1:500
Insulin	Mouse	Novo Nordisk	1:100
Insulin	Guinea Pig	Abcam	1:500
Somatostatin	Sheep	Greg Aponte	1:500
Somatostatin	Mouse	Novo Nordisk	1:100
Pancreatic Polypeptide	Guinea Pig	DAKO	1:1000
PYY	Rabbit	Novo Nordisk	1:1000
CART	Rabbit	Novo Nordisk	1:1000
Ghrelin	Rabbit	C. Tomasetto	1:500
Pdx-1	Goat	BCBC	1:5000
Pdx-1-253	Rabbit	Joel Habener	1:20,000
Nkx6.1	Mouse	BCBC	1:4000
Nkx 2.2	Mouse	DSHB	1:2000
MafA	Rabbit	Bethyl lab	1:1000
MafB	Rabbit	Bethyl lab	1:1000

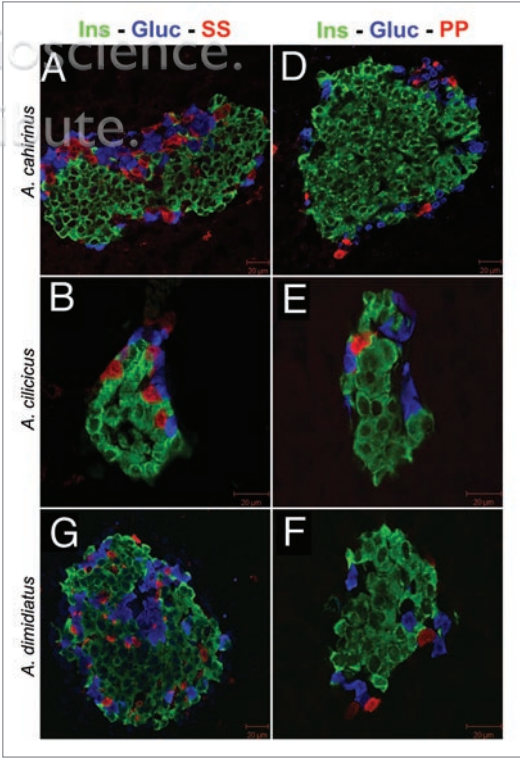


Figure 2. Immunoreactivity for insulin, glucagon, somatostatin and pancreatic polypeptide. (A–C) Immunoreactivity for insulin (green), glucagon (blue) and somatostatin (red) in three species of *Acomys*. (D–F) Immunoreactivity for insulin (green), glucagon (blue) and pancreatic polypeptide (red) in three species of *Acomys*. Scale bar = 20 μ m.

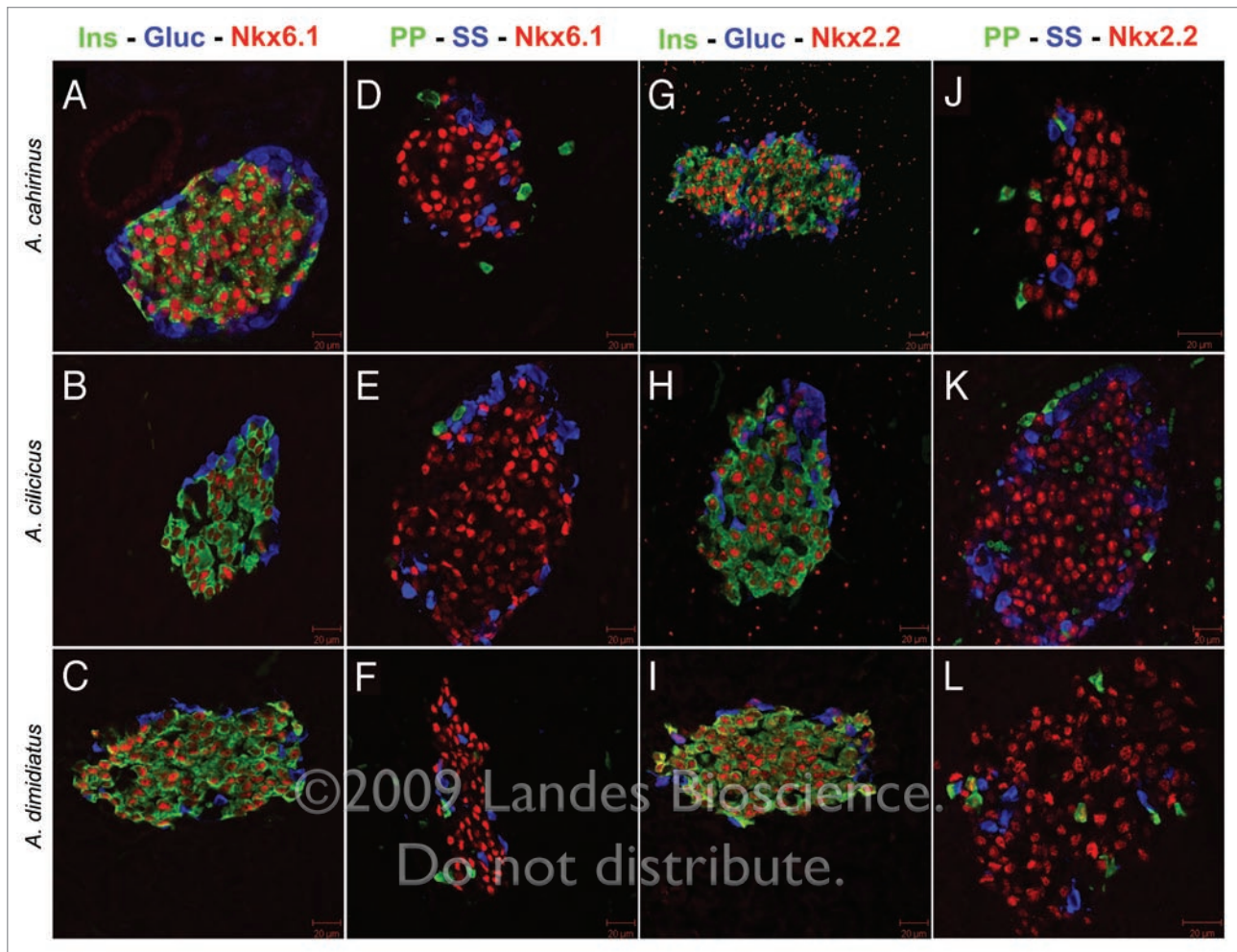


Figure 3. Localization of the Nkx family transcription factors. (A–C) Immunoreactivity for insulin (green), glucagon (blue) and Nkx6.1 (red) in three species of *Acomys*. (D–F) Immunoreactivity for pancreatic polypeptide (green), somatostatin (blue) and Nkx6.1 (red) in three species of *Acomys*. (G–I) Immunoreactivity for insulin (green), glucagon (blue) and Nkx2.2 (red) in three species of *Acomys*. (J–L) Immunoreactivity for pancreatic polypeptide (green), somatostatin (blue) and Nkx2.2 (red) in three species of *Acomys*. Scale bar = 20 μ m.

maturity-onset diabetes of the young and type 2 diabetes; absence of the gene leads to complete pancreas agenesis.^{11,12} Pdx-1 in the adult pancreas is normally expressed in all the β - and δ -cells. Interestingly, Leibowitz et al.²⁷ and Vedtofte et al.²⁸ have shown the apparent absence of this crucial factor in the sand rat (*Psammomys obesus*). In the same subfamily (Gerbillinae), we have observed that five species of the genus *Meriones* demonstrate the same absence of Pdx-1 protein and 13 species lack the gene.¹³ In the genus *Acomys* the closest relatives to Gerbillinae, we can demonstrate that the Pdx1 protein is immunocytochemically detectable in the expected cell types in all three *Acomys* species. Cloning of the gene confirms its high similarity to other rodent species. Uniquely, we found that the sequence of five glycines found in the rat (amino acids 216–220) was extended to eight in the *Acomys cahirinus* species. Additionally, 5 of the 16 amino acid changes were to proline residues which are known to effect the folding of proteins. The molecular consequences of these changes will require further study.

This is perhaps an unexpected finding as spiny mice have been used as models in the study of type 2 diabetes and shown to have defective insulin secretion.²⁹ The closely related gerbil, *P. obesus*, gains weight and develops hyperglycemia and hyperinsulinemia when placed in the lab and feed on normal rat chow which is a high energy (HE) diet compared to their native diet, the salt bush. The spiny mice kept on HE diet also gain weight, but show different characteristics in the development of diabetes.³⁰ Their weight gain is associated with a large growth of the pancreatic islets and increased insulin content, while the ability to secrete insulin in response to glucose is weak.¹⁸ These characteristics are similar to those observed in *Psammomys obesus*. This suggested that perhaps *Acomys*, like their close relatives, *P. obesus*, might also lack Pdx-1. This did not turn out to be the case. What causes the defective insulin secretion in *Acomys* is not related to lack of Pdx-1 protein as maybe the case in *P. obesus*.^{27,28} Furthermore, based on genetic data, it has been argued that the spiny mouse (*Acomys*) is more closely related to gerbils (Gerbillinae) than to real mice (Murinae).¹⁵ In addition, we have recently shown that

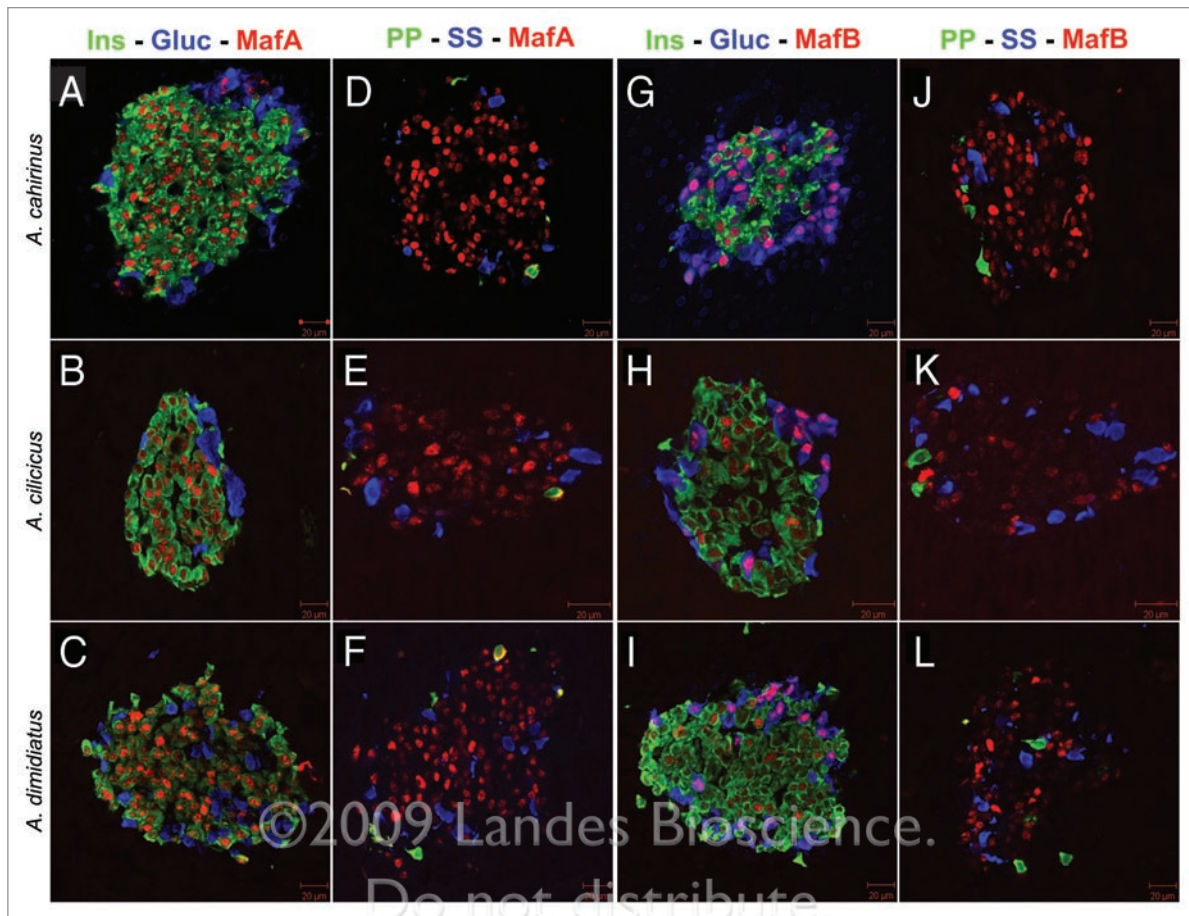


Figure 4. Localization of the MafA and MafB transcription factors. Immunoreactivity for insulin (green), glucagon (blue) and MafA (red) in three species of *Acomys*. (D–F) Immunoreactivity for pancreatic polypeptide (green), somatostatin (blue) and MafA (red) in three species of *Acomys*. (G–I) Immunoreactivity for insulin (green), glucagon (blue) and MafB (red) in three species of *Acomys*. (J–L) Immunoreactivity for pancreatic polypeptide (green), somatostatin (blue) and MafB (red) in three species of *Acomys*. Scale bar = 20 μ m.

in the subfamily Otomyinae, the species *Otomys sloggetti robertsi* has a readily detectable Pdx-1 expression.¹³ Thus, all subfamilies of Murinae except Gerbillinae have readily detectable Pdx-1 protein in their islets.

Morphologically, the islets of the three *Acomys* species examined in this study were similar. In general, the islets were round, oval or oblong with insulin immunoreactive β -cells being the major cell type, positioned in the central part of the islets and glucagon, SS or PP immunoreactive cells scattered in the rims of the islets, with glucagon being the most abundant of the three. These findings are in line with observations in other rodents so far examined (*Rattus*, *Mus*, vole), except for the African Ice Rat (*Otomys sloggetti robertsi*), where the glucagon producing α -cells were found to be the dominant cell type in some islets.¹⁴

PYY has been reported to be localized in the endocrine cells of the distal intestine but also in non-intestinal tissues like in endocrine cells of the stomach and the pancreas.^{31,32} We found PYY to be expressed in all three species of *Acomys*. It was generally found not to co-express other hormones but occasionally co-express PP. This is similar to what we observed in the African Ice Rat, although sometimes cells are found to co-express glucagon.¹⁹ Rats have been shown to express PYY in α , δ or PP cells.²⁵ Thus,

PYY can be observed to co-localize with most endocrine cells in the islet except for of insulin.

We failed to detect any CART or ghrelin expressing cells in all three *Acomys* species. Whether this is due to the antiserum or if these hormones are not expressed in these animals, is unknown. The antisera worked on control mouse sections ruling out a failure of the antisera to recognize the protein. Access to other *Acomys* tissues where these peptides are expressed could help us to determine if it is lack of expression or the divergence of the peptide structures. Ghrelin cells are not commonly found in other adult rodents either,^{7,14} while CART is normally co-localized with somatostatin.⁶

The transcription factor Islet 1 (*Isl1*) is expressed in all the endocrine cell types of the pancreas.³³ We observed the same phenomenon in the three studied spiny mice species. The level of expression varied among the different cell types but tended to be highest in the β -cells. Members of the NK2 family of homeodomain transcription factors, *Nkx6.1* and *Nkx2.2*, are regulators of the differentiation of pancreatic endocrine cells.³⁴ During development, the transcription factor *Nkx6.1* is expressed in the developing foregut and in the central nervous system.³⁵ In the adult pancreas, *Nkx6.1* becomes restricted to be expressed

in the pancreatic β -cells.³⁶ We observed this expression pattern also in the pancreas of the spiny mice. Nkx6.1 is found only in the β -cells. However, Nkx2.2 is expressed early in developing pancreatic buds and is later restricted to α -, β - and PP cells of the islets, which is also what we have detected.

We also analyzed other transcription factors important for the function of the pancreas. MafA was reported to be a glucose regulated, β -cell specific transcriptional activator of the insulin gene.^{37,38} We found MafA to be expressed in the insulin immunoreactive cells, although at variable levels with some cells having little or no MafA. Interestingly, we also observed MafA immunoreactivity in the cytoplasm of PP-cells, which we also detected in five species of *Meriones* gerbils.¹³ Thus, MafA or a protein recognized by the MafA antiserum is found in the cytoplasm of pancreatic polypeptide cells. This phenomenon is not detected in mice and is accordingly an unusual finding. We used two different MafA antisera and run cross reactivity controls as well as absorption experiments, but this did not explain the finding. MafB was found to be expressed in the adult mouse pancreas solely in the α -cells.³⁹ In the *Acomys* species, MafB was observed at variable levels in the α -cells, but the protein was also found to be expressed in the β -cells at somewhat equal levels. The same observation was recently reported in the study of the endocrine pancreas of the African Ice rat *Otomys sloggetti robertsi*.¹⁴ Neither PP nor somatostatin immunoreactive cells co-expressed MafB. During development of the pancreas in mice, the two proteins are co-expressed but later they separate to be expressed in α - and β -cells,^{39,40} which might suggest that in these desert rodents some β -cells resemble a more embryonic phenotype.

In conclusion, *Acomys*, the most closely related species to the gerbils, have the Pdx-1 gene and express the protein. This suggests that during the evolution, when the gerbils and spiny mice diverged, the need for the Pdx-1 gene was lost. As Pdx-1 was easily detectable, this can not likely explain the poor responsiveness of *Acomys* β -cells to glucose. Finally, the regional distribution of the endocrine cells in the pancreatic islets of *Acomys cabirinus*, *A. cilicicus* and *A. dimidiatus* was found to be similar to that described in other rodent species, except for the expression pattern of MafA and MafB.

Materials and Methods

Animals. *Acomys cilicicus* and *Acomys dimidiatus* were obtained from a colony maintained at the Institute of Parasitology, Academy of Sciences of the Czech Republic, Ceske Budejovice. *Acomys cabirinus* were obtained from a colony maintained at Monash University, Australia. Animal handling and sacrifice were carried out according to regulations specified under the Protection of Animals Act by the Authority in Denmark and Novo Nordisk A/S. All *Acomys* experiments were approved in advance by Monash University School of Biomedical Sciences Animal Ethics Committees and conducted in accordance with

the Australia Code of Practice for the Care and Use of Animals for Scientific Purposes. Pancreatic tissue was fixed in either 4% paraformaldehyde or 10% saline buffered formalin overnight and transferred to 70% ethanol before paraffin embedding.

Immunohistochemistry. Pancreatic tissue embedded in paraffin were cut in 4 μ m sections using the 2055 Autocut microtome from Leica Instruments GmbH (Wetzlar, Germany) and dried overnight at 37°C. Deparaffination was done in xylene followed by rehydration in graded alcohol solutions. Endogenous peroxidase activity was blocked with 3% hydrogen peroxide for 5 minutes followed by a brief rinse in water before microwave oven treatment in Tris-EDTA pH 9.0 (650 watts for 4 minutes followed by 160 W for 15 minutes). Nonspecific binding was blocked with 10% normal donkey serum for 10 minutes followed by TNB blocking buffer (0.1 M Tris-HCl, pH 7.5, 0.15 M NaCl, 0.5% blocking reagent) for 30 minutes [for Tyramide Amplification (TSA) staining only] and incubated with a combination of two or three primary antibodies overnight at room temperature. The subsequent day, the TSA sections were incubated with donkey anti-rabbit or -mouse secondary biotinylated antibody (Jackson ImmunoResearch Laboratories; West Chester, PA) for 30 minutes and horseradish peroxidase-streptavidin conjugate (Zymed/Invitrogen; Taastrup, Denmark) for 15 minutes. Visualization of one of the antibodies was developed with TSA substrate (Cy3, 1:100) (Perkin Elmer; Hvidovre, Denmark) for 10 minutes. The one or two other primary antibodies (Table 2) were visualized with Cy2 (1:500) and Cy5 (1:500) highly cross absorbed secondary fluorescent antibodies for multiple labeling (Jackson ImmunoResearch). Between each step, the sections were rinsed in phosphate buffered saline (PBS). Negative control sections were run in parallel omitting the addition of primary antibodies. In addition, cross reactivity of the secondary antisera were tested, and no non-specific staining or cross reactivity of our highly cross absorbed secondary antisera was observed (data not shown). Pictures were taken using Zeiss LSM 510 META scanning laser microscope (Carl Zeiss; Oberkochen, Germany) and the figures were produced in Canvas 8.0 (ACD Systems; Victoria, British Columbia, Canada).

Cloning of the Pdx-1 gene. PCR was performed on genomic DNA or cDNA prepared from pancreatic RNA. The PCR primers were directed to the conserved 5' end of the Pdx-1 gene and later the rest of the gene.¹³ The PCR primer sequences are available upon request. The PCR product was cloned using TA cloning following the manufacturers suggestions (Invitrogen). Sequencing was performed by MWG Biotech (Germany) and alignments performed in Vector NTI 9.0 software. The sequence of *A. cabirinus* Pdx-1 gene has been deposited in the GenBank database with accession number GQ179992.

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